

CMSAPI v3

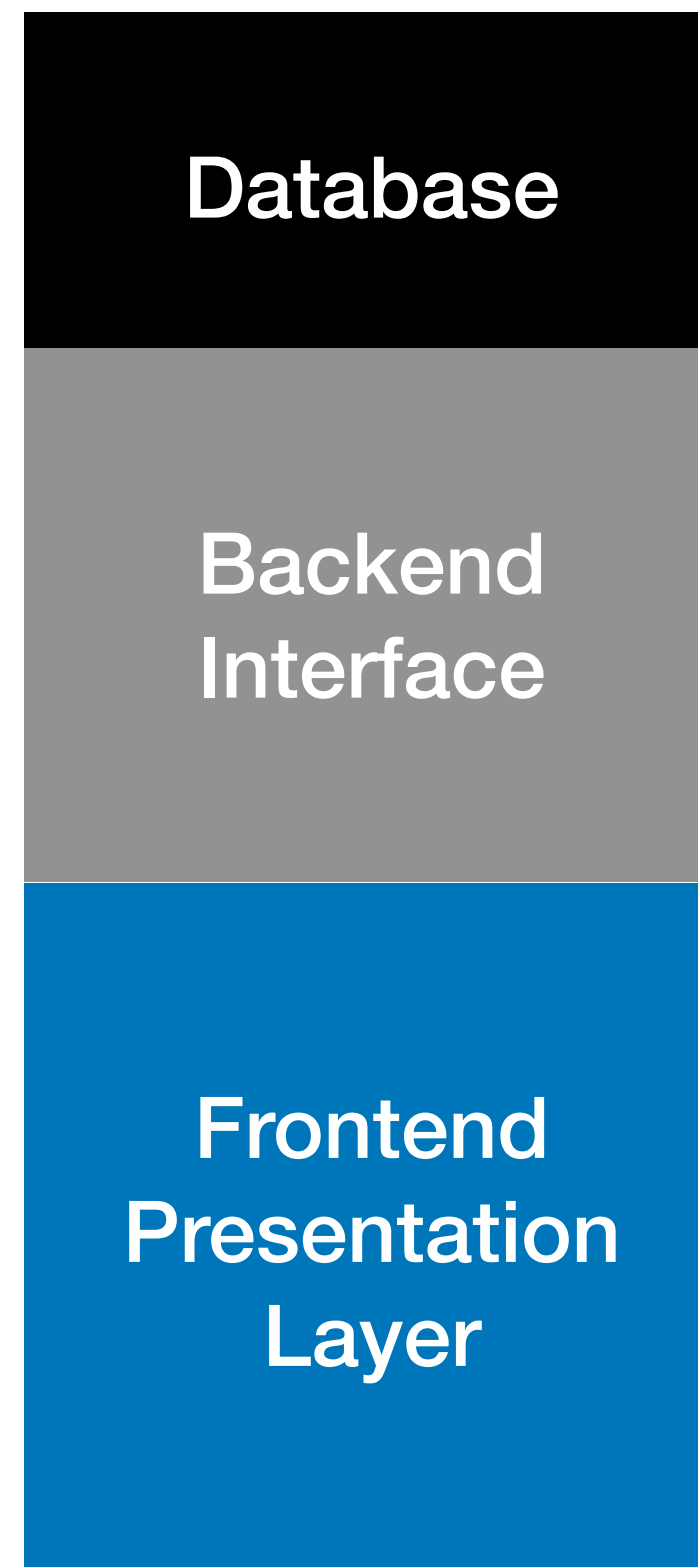
**Python-based REST API
to retrieve, analyze and consolidate data**

Decoupled CMS Architecture • FastAPI as Gateway/Middleware

<https://api.cms.test.unibe.ch/v3>

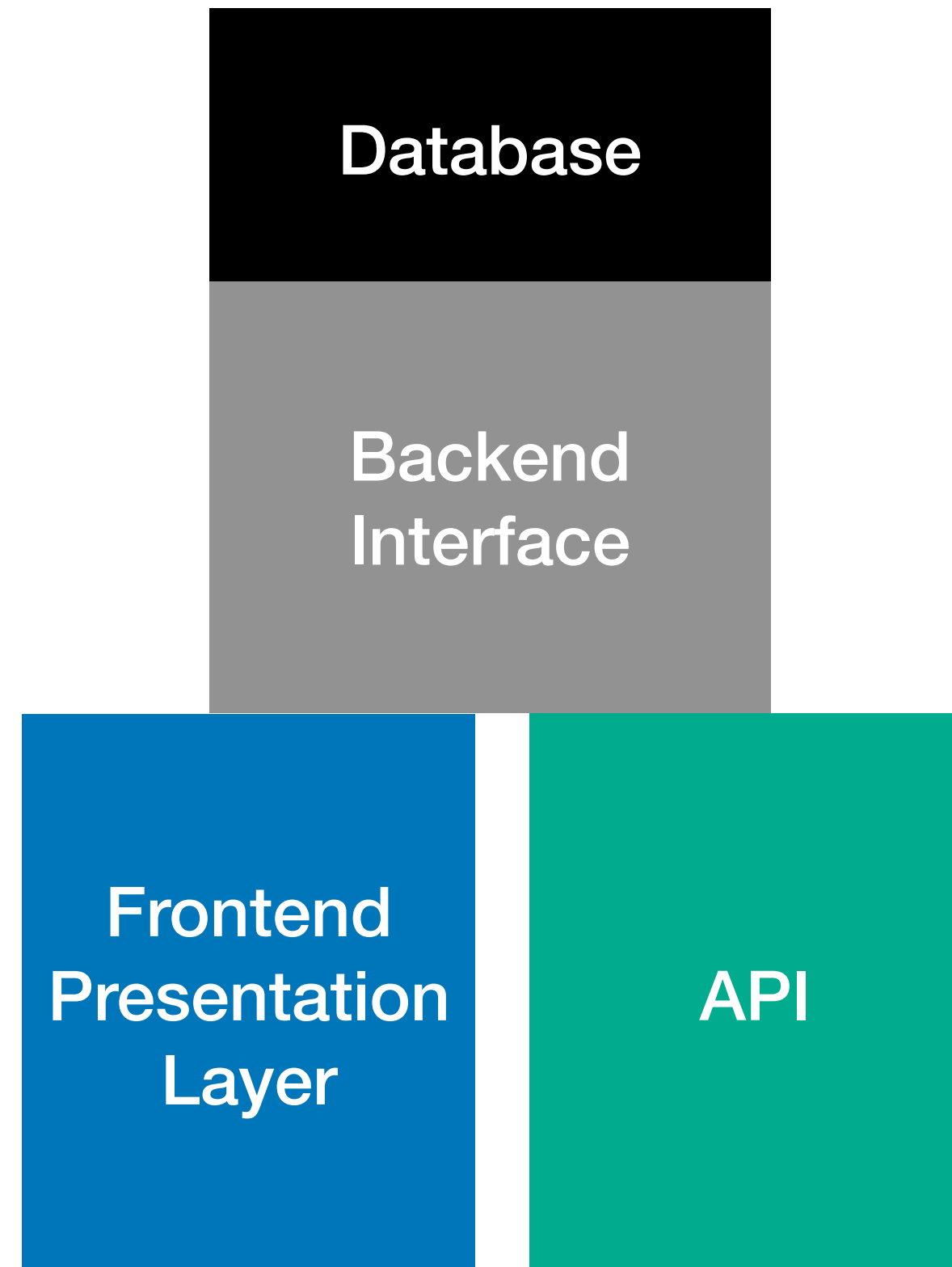
Christian Meier, 19.01.2023
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Traditional CMS



Backend and Frontend on the same server

Decoupled CMS



Headless CMS

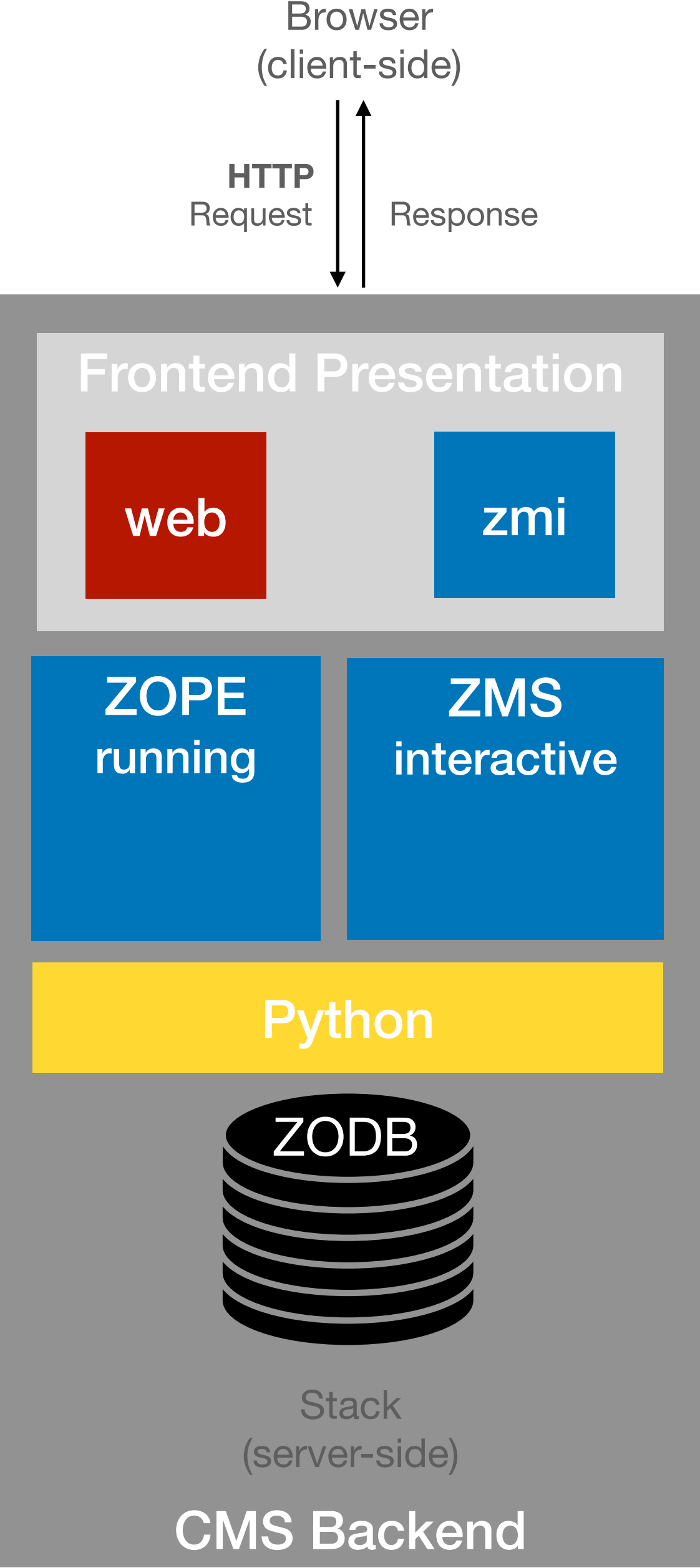


Backend accessible from multiple platforms via an API

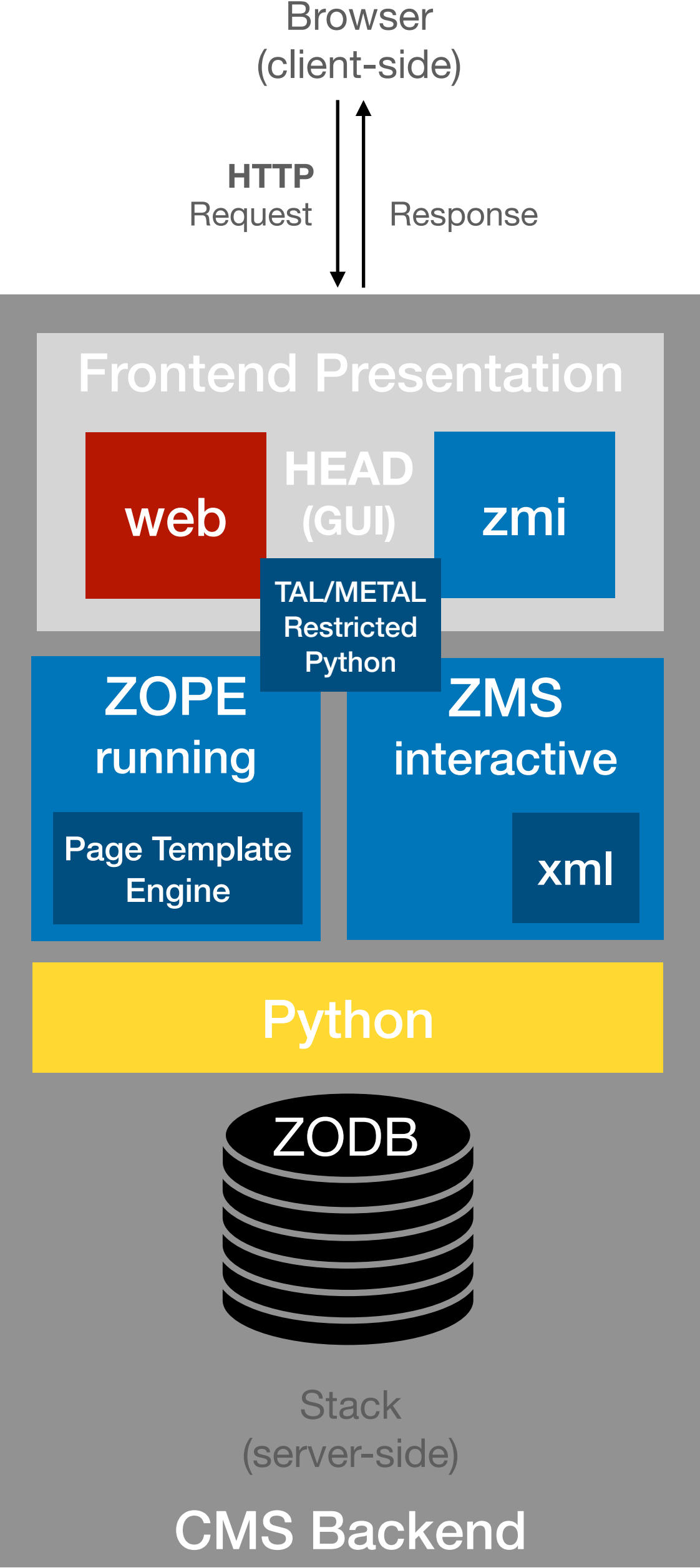
Traditional CMS



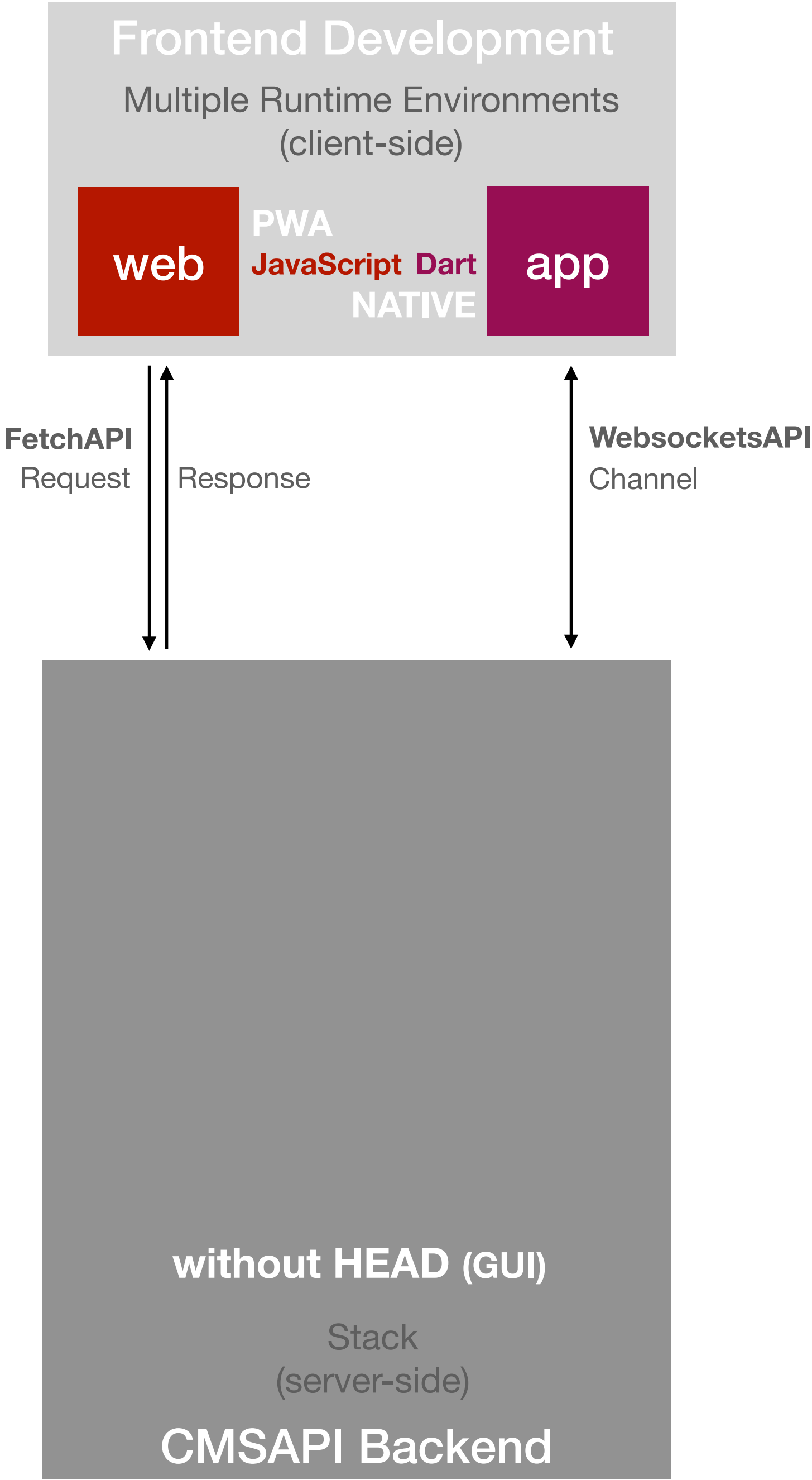
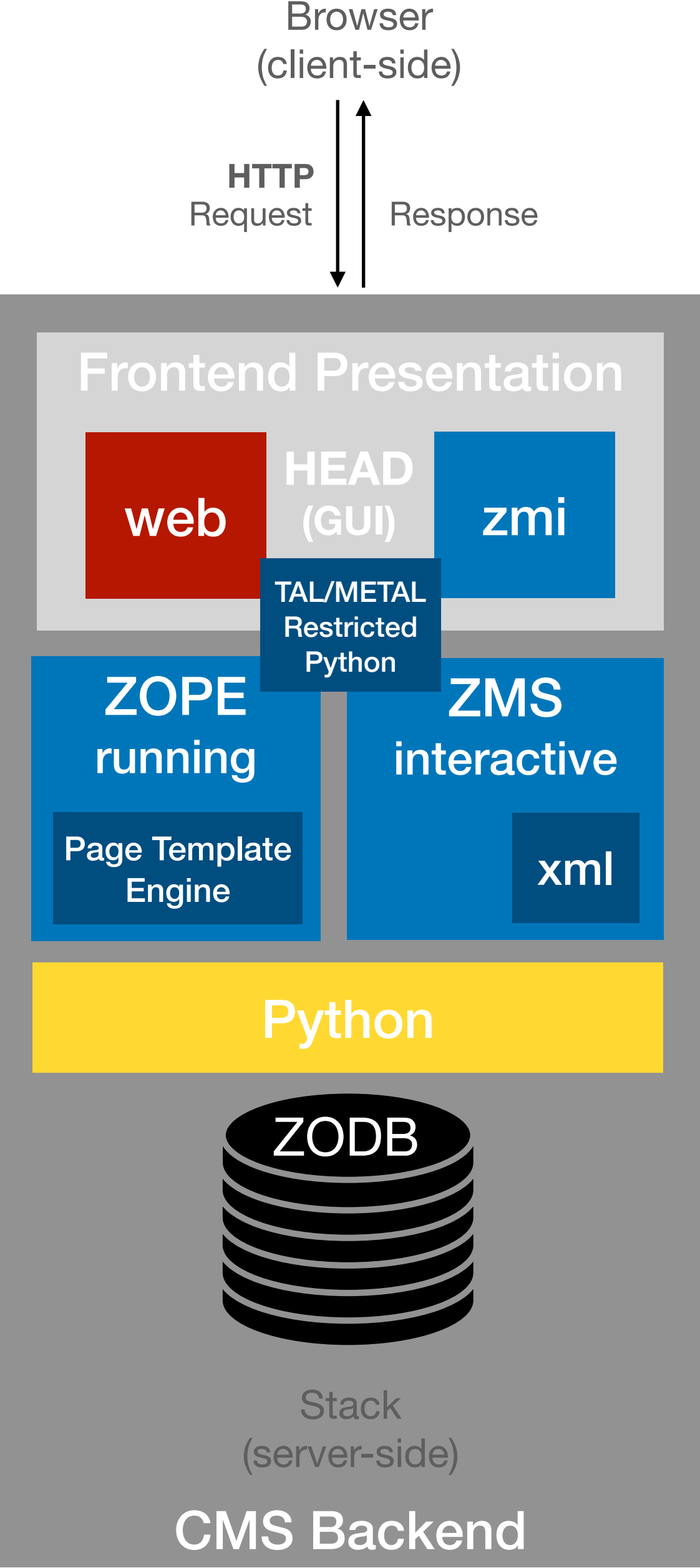
Traditional CMS



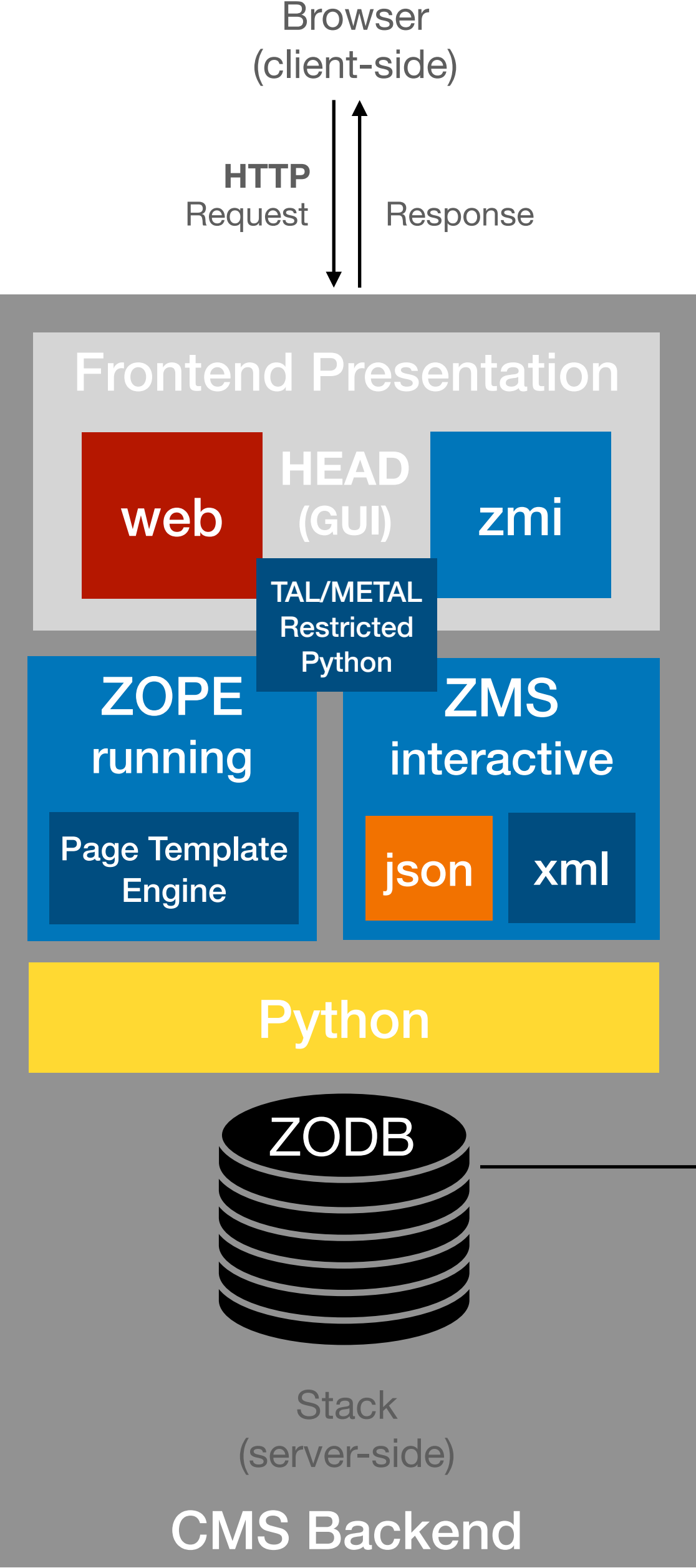
Traditional CMS



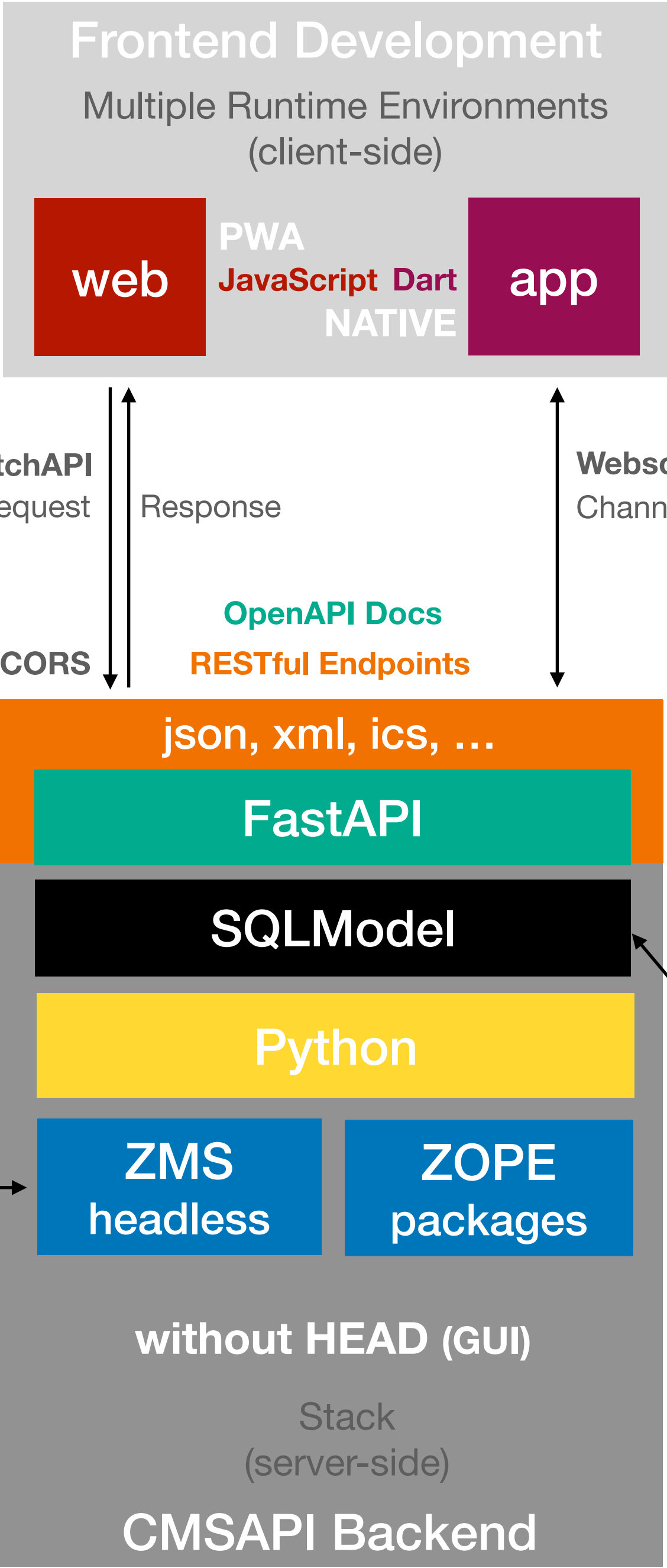
Decoupled CMS



Decoupled CMS

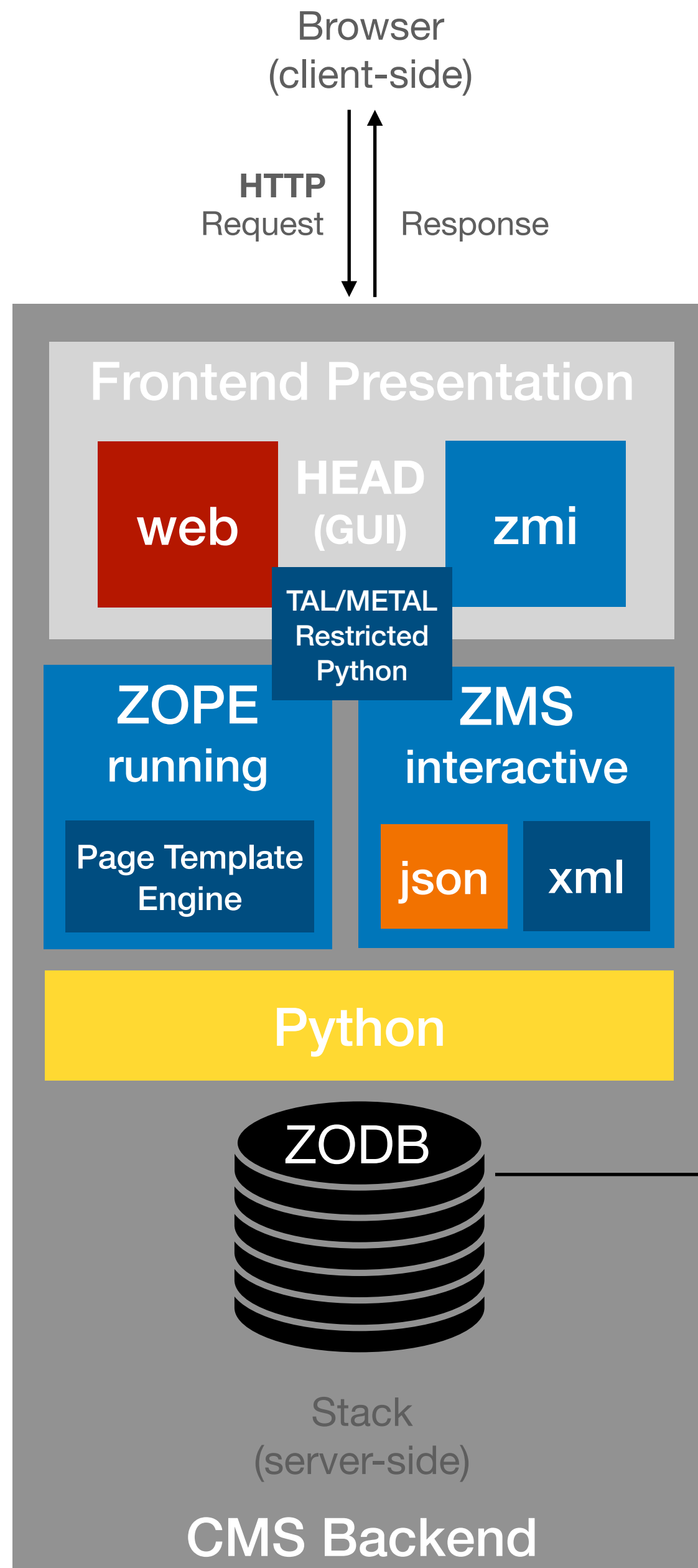


zeo://
read-only

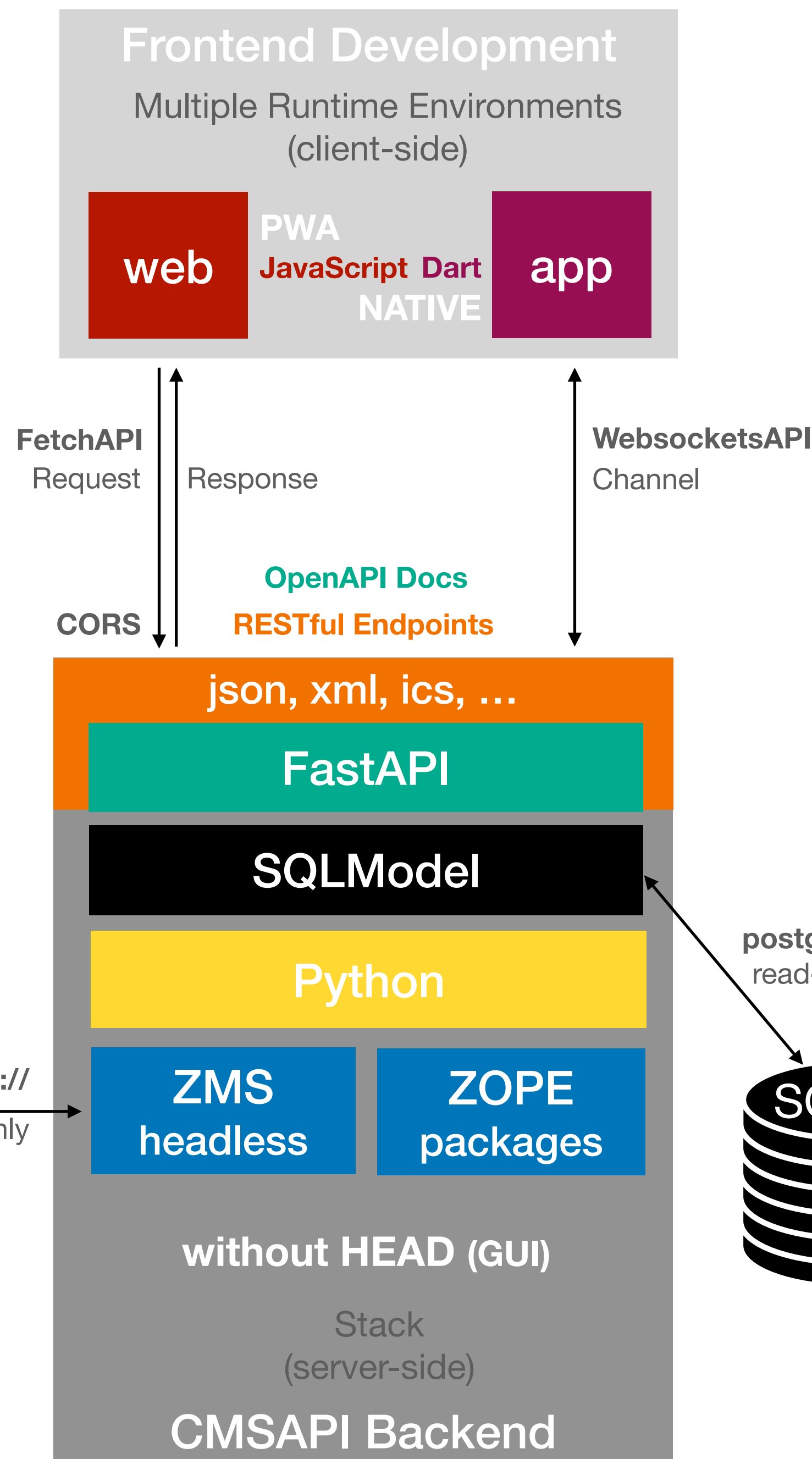


postgres://
read-write

Decoupled CMS



zeo://
read-only



postgres://
read-write

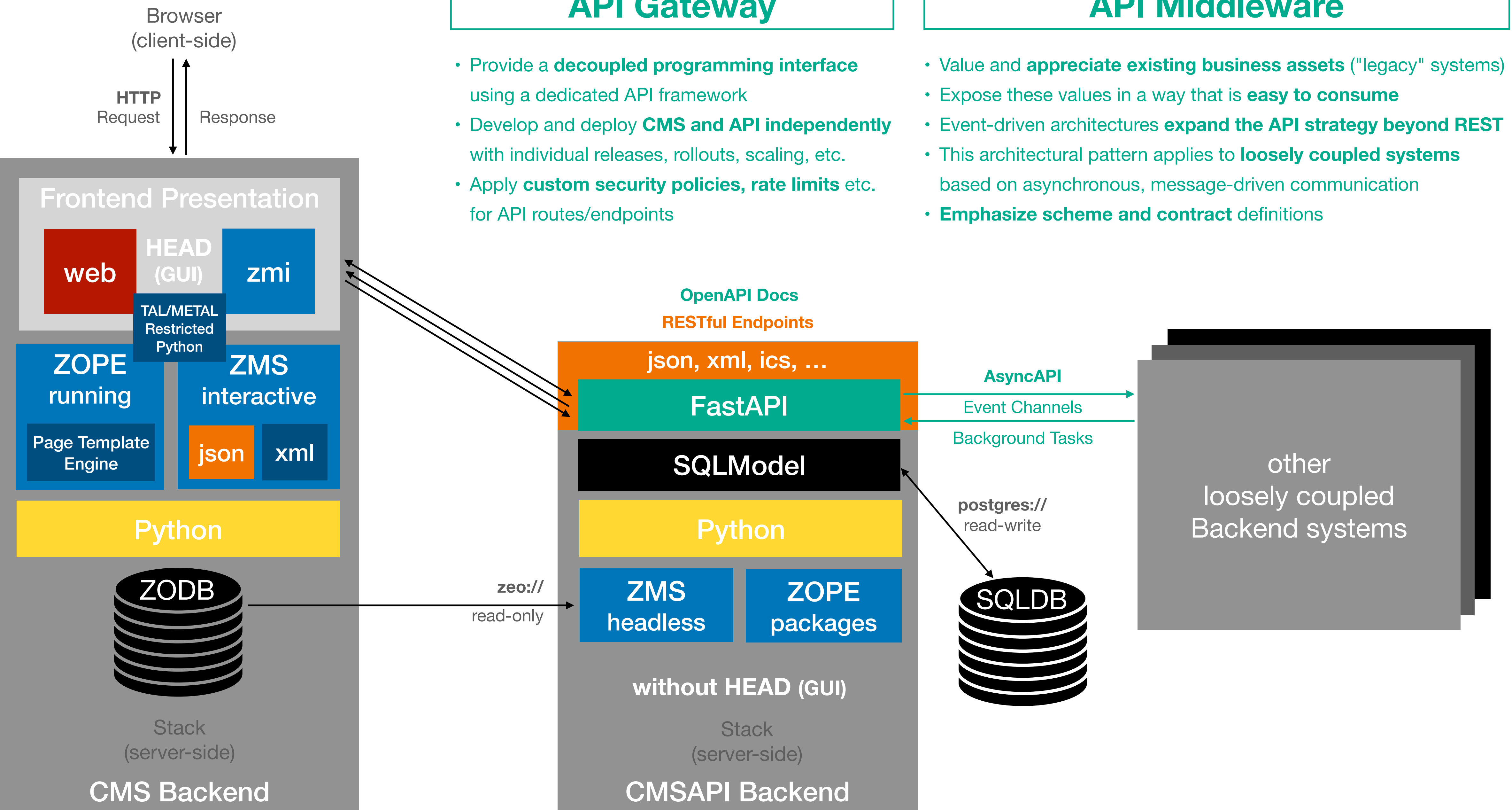


FastAPI is a modern web framework for building APIs with Python 3.7+ based on standard Python type hints, JSON schema, and OpenAPI (formerly known as Swagger).

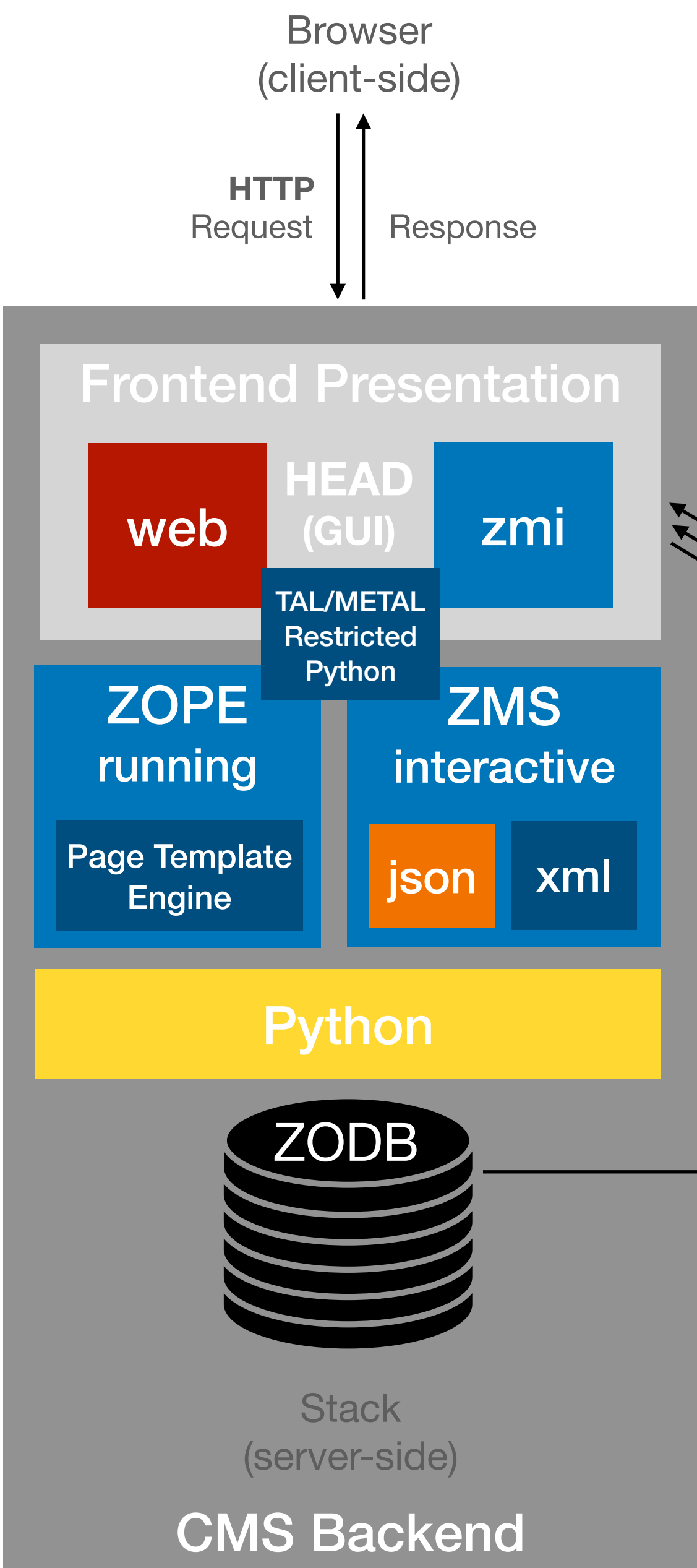
The key features are:

- **Fast in use:** Very high performance, on par with NodeJS and Go (thanks to Starlette and Pydantic).
- **Fast to code:** Great IDE/Editor support. Code completion everywhere. Minimize code duplication. Less time debugging. Automatic interactive documentation.
- **Easy to learn:** Designed to be easy to use and learn. Less time reading docs. Many tutorials and howtos available.

Decoupled CMS



Decoupled CMS

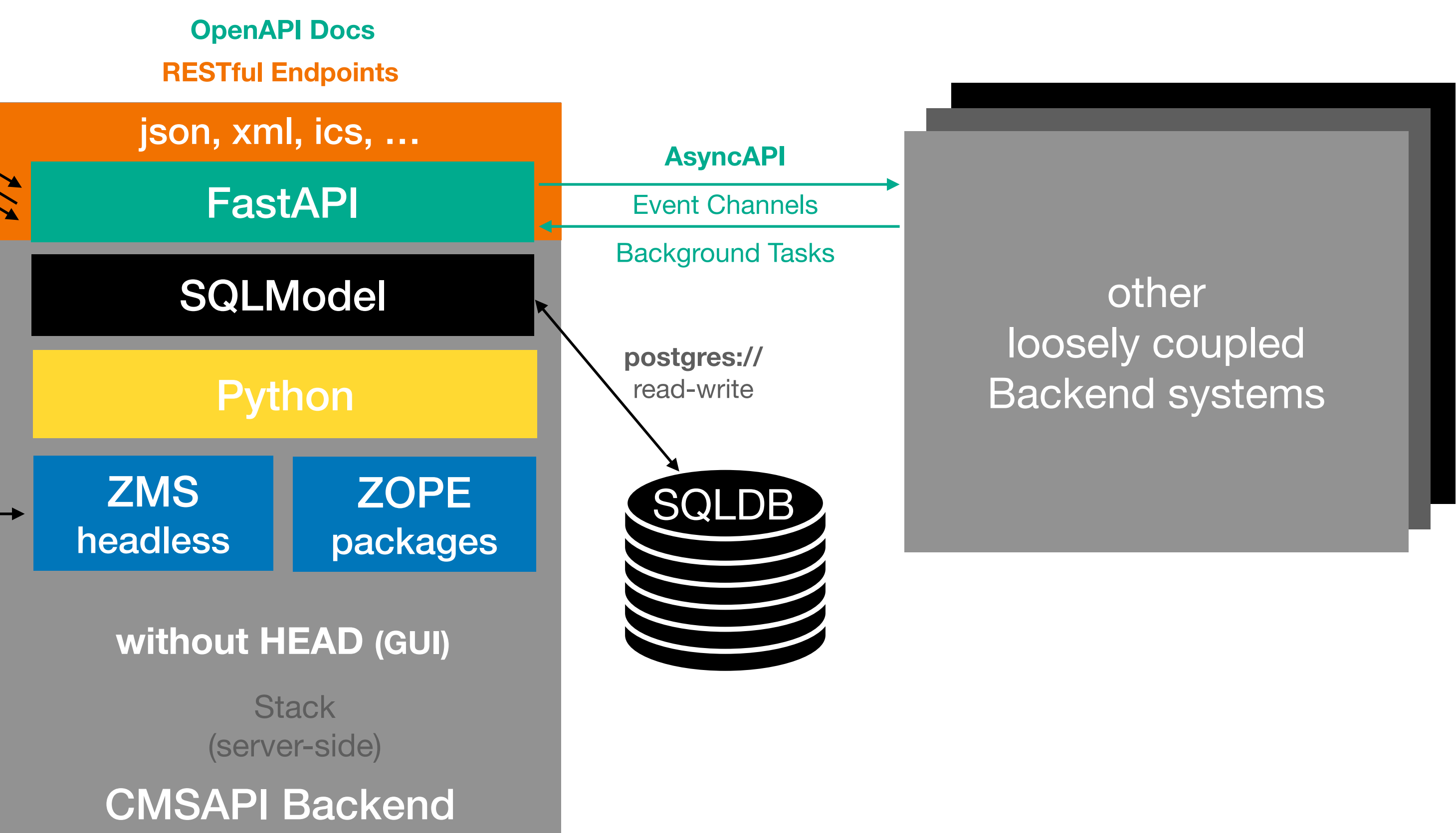


Benefits

- flexible, adaptive and scalable by design
- minimize upstream dependencies in the stack caused by legacy
- leverage best in class modern frameworks and paradigms
- developer resources for modern stacks may easier to recruit than for legacy

Drawbacks

- more complex infrastructure must be managed
- adds an additional abstraction layer that has to be developed and maintained
- can be a rather volatile and experimental environment
- needs potentially more developer know-how and capacity to take advantage of it



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